

JILL WANG

Data Scientist & Software Developer

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Kaggle: <https://www.kaggle.com/jillwang87> · **Portfolio Site:** <https://jillwang87.github.io>

WORK EXPERIENCE

Head of Analytics

Jul 2023 – Present

Truthifi - digital investment monitoring platform

- Integrated and blended data from APIs of vendors such as Xignite, Morningstar, Yahoo Finance, public data from the government such as from the SEC, the Municipal Securities Rulemaking Board (MSRB) and custom web-scraped sources
- Designed advanced algorithms leveraging regex matching and natural language processing techniques to accurately reconcile securities across sources with inconsistent or incomplete unique identifiers
- Engineered robust data-cleaning solutions to meticulously standardize, validate, and enhance the quality of market data, ensuring precision and reliability for downstream analysis
- Collaborated cross-functionally with product and engineering teams to integrate data insights into a seamless customer platform experience
- Developed real-time performance feedback and diagnostic tools to benchmark investment portfolios against comprehensive market data
- Technology: TypeScript, MongoDB, AWS

Data Scientist (via Adastra)

Mar 2023 – Jun 2023

Mark Anthony Group - Alcoholic beverage company with iconic products such as White Claw Hard Seltzer, Mike's Hard Lemonade

- Built a sales forecasting Machine Learning (ML) model for the purpose of a gap and attention marketing analysis (Gap: identify products that are not sold in a store, but should be sold; Attention: identify products that are not selling well and require attention)
- Utilize PySpark to build ML pipelines to process millions of rows of data
- Technology: Python, AWS SageMaker, XGBRegressor, SQL

Operation Research Analyst (via Adastra)

Aug 2022 – Feb 2023

Taiga Building Products Ltd. - Wholesale distributor of building materials

- Built a system to optimize distribution routes and truck-loads, reducing freight costs and improving sales margins
- Built an address-clustering algorithm to cluster customers' geographic locations and optimize delivery routes using a clustering AI model and Google Direction API
- Built a truck-load optimization model by mathematical formulation using Google OR
- Technology: Python, Azure ML, Google OR-Tools, Google Direction API

Data Analytics Developer (via Adastra)

May 2021 – Jun 2022

TD Bank – This role is part of the project delivery team

- Built a Python-based data lifecycle management (DLM) application that uses Computational Linguistics and Natural Language Processing (NLP) to extract features for a machine learning pipeline which predicts security classifications and retention periods

- Built a user-friendly interface with tkinter for business users to provide a location (either local files or a Hadoop database) to scan and automatically run the ML pipeline, emitting the predictions in an Excel file
- The newly automated process delivers predictions in minutes that previously took months of manual work from data stewards
- ETL from Hadoop using Impala and developing Tableau dashboards to present insights to business executives
- Technology: Impala, Python, scikit-learn, NLP, tkinter, spaCy, Tableau, Oozie

Data Science Consultant

Apr 2021 – Jun 2023

Adastra North America - Data and AI consultancy

- Extensive training in leveraging cloud platforms to design and implement scalable analytics and machine learning solutions
- Assigned to TD Bank, Taiga Building Products Ltd. and Mark Anthony Group as listed above

Data Scientist & Software Developer (Intern)

Apr 2019 – Aug 2019

Graphen – R&D Group developing anti-money laundering platform

- Developed an unsupervised model on time-series data and a supervised model on multivariate data, both for the purpose of performing financial behaviour anomaly detection/fraud detection
- Applied statistical methods to design a systematic way to calculate a risk (anomaly) score
- Applied a numerical analysis method (power spectral density) to improve the accuracy of a time series model
- Developed the front end of a React-based anti-money-laundering platform
- Technologies: Python, NumPy, pandas, scikit-learn, XGBoost, Git, JavaScript, React, SASS, CSS3

PROGRAMMING LANGUAGES & TOOLS

Data Science: Python · R · MATLAB

Data Engineering: SQL · MongoDB · Hadoop

Business intelligence: Tableau

Software Development: JavaScript · TypeScript · React · Git

Education

Honours Bachelor of Mathematics in Computational Mathematics

University of Waterloo, 2021

Honours Bachelor of Mathematics in Statistics, Computing Technology Option

University of Waterloo, 2021

Education Relevant Coursework

Data Science: Statistical Learning-Classification · Sampling and Experimental Design · Generalized Linear Models & Applications · Applied Probability · Data Visualization · Computational Statistics and Data Analysis · Mathematical Statistics

Applied Math: Applied Cryptography · Deterministic Operation Research Models · Computational Methods for Differential Equations

Computer Science: Data Types and Structures · Elementary Algorithm Design and Data Abstraction · Object-Oriented Software Development